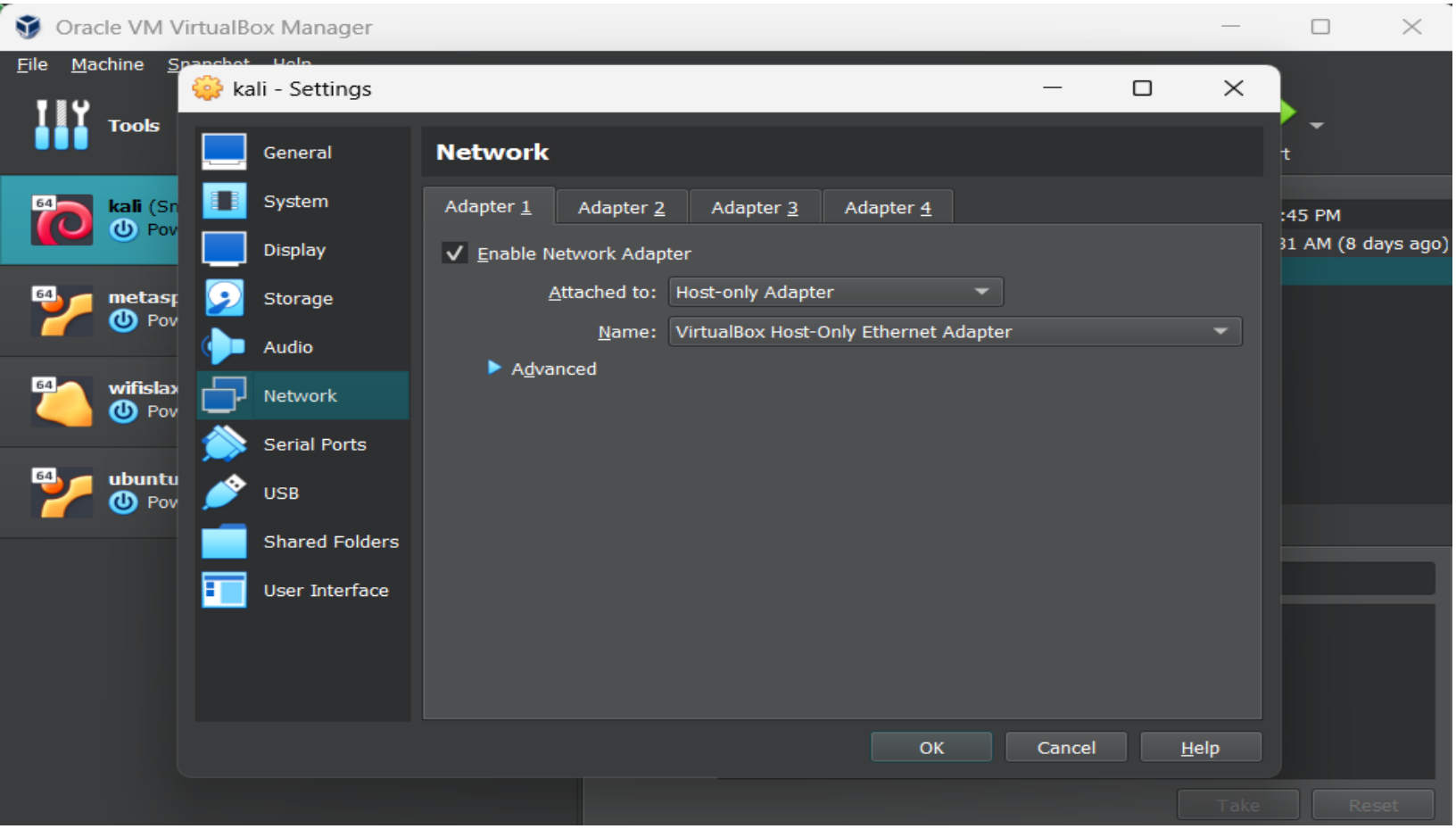
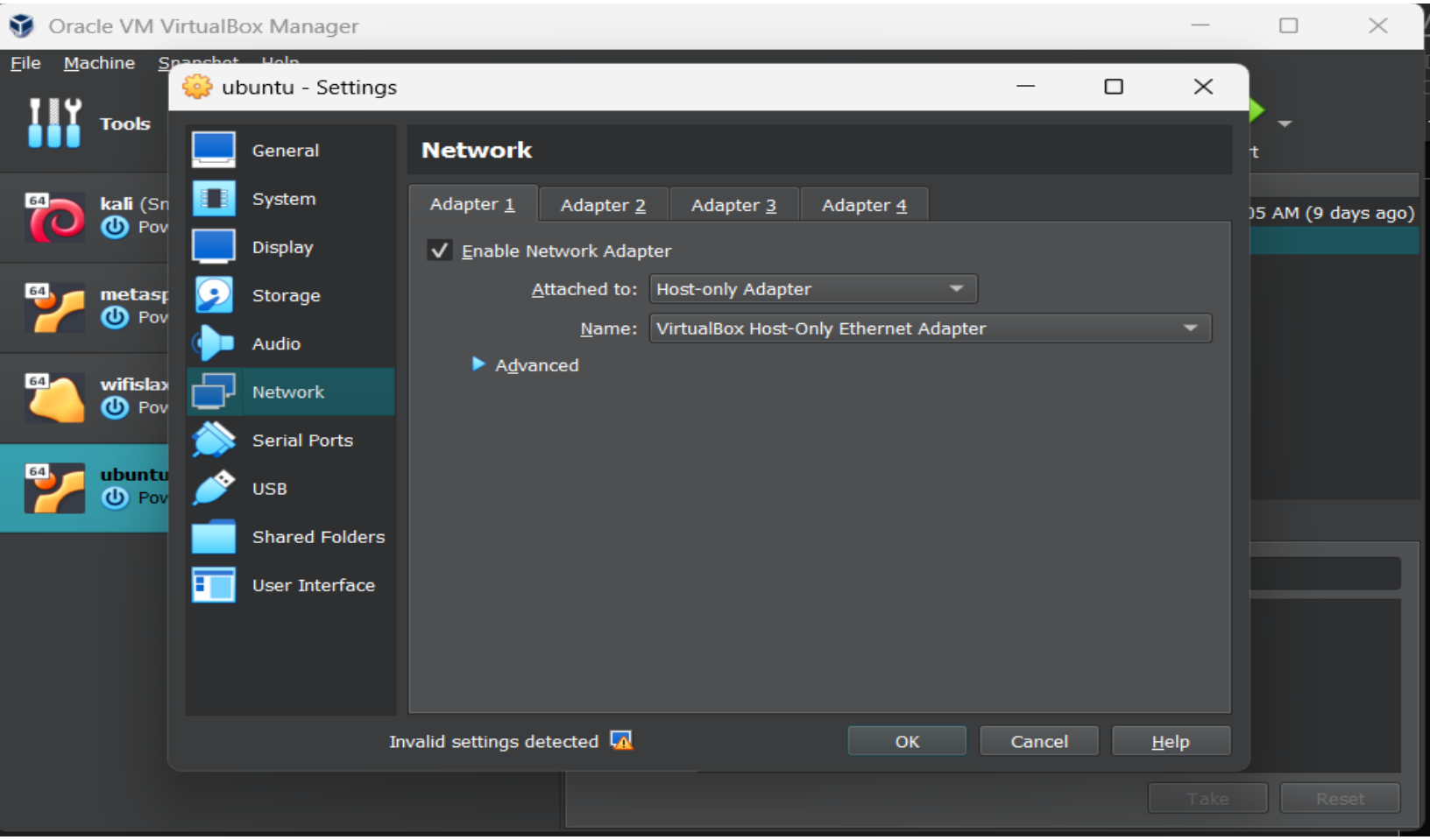
NETWORK MONITORING AND ATTACK DETECTING (Snort)

Objective: Monitor and detect network-based attacks using IDS or traffic analyzers.

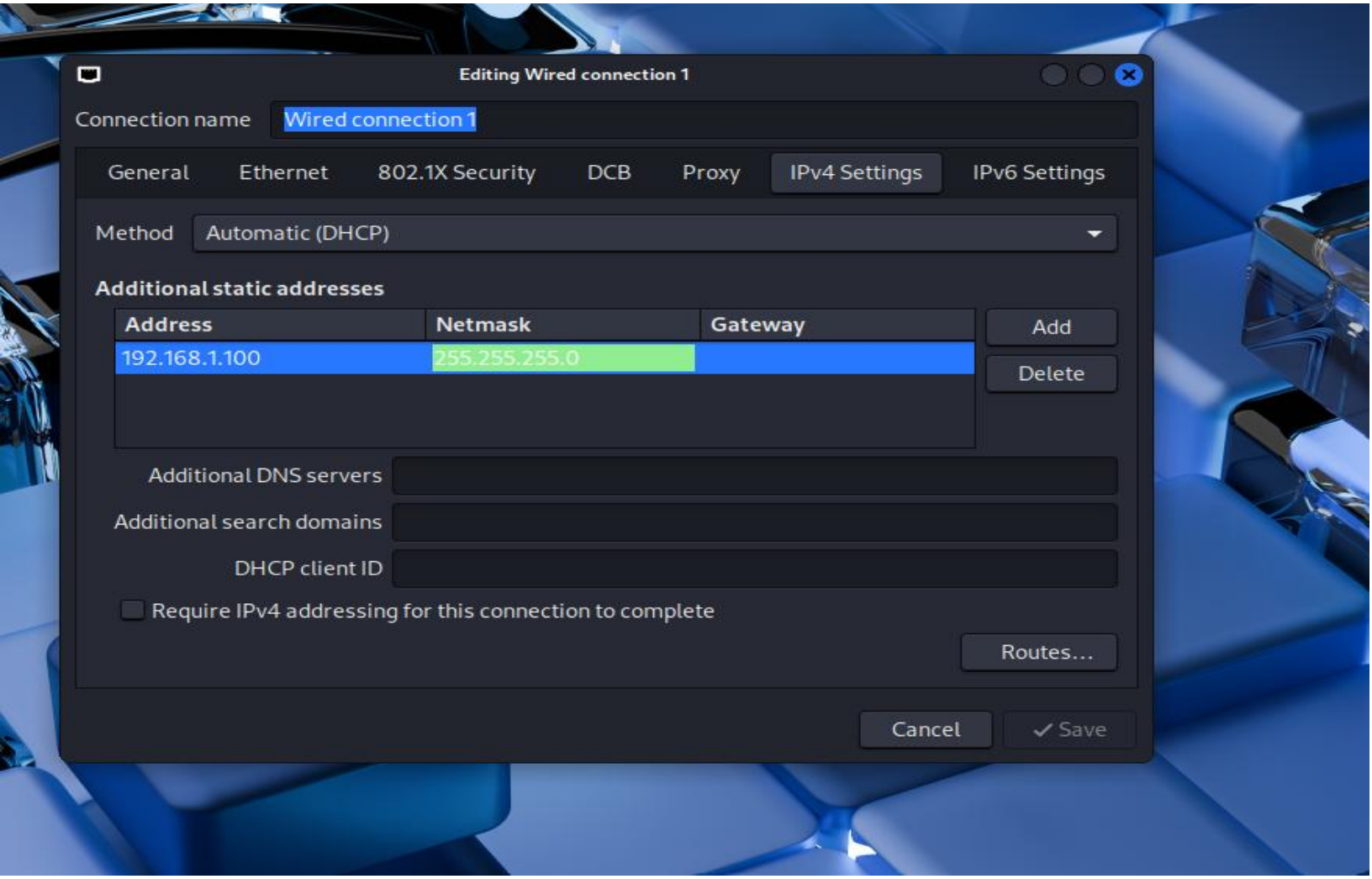
Instructions:

- Use Snort
- Simulate common attacks: port scans, DoS, ARP spoofing.

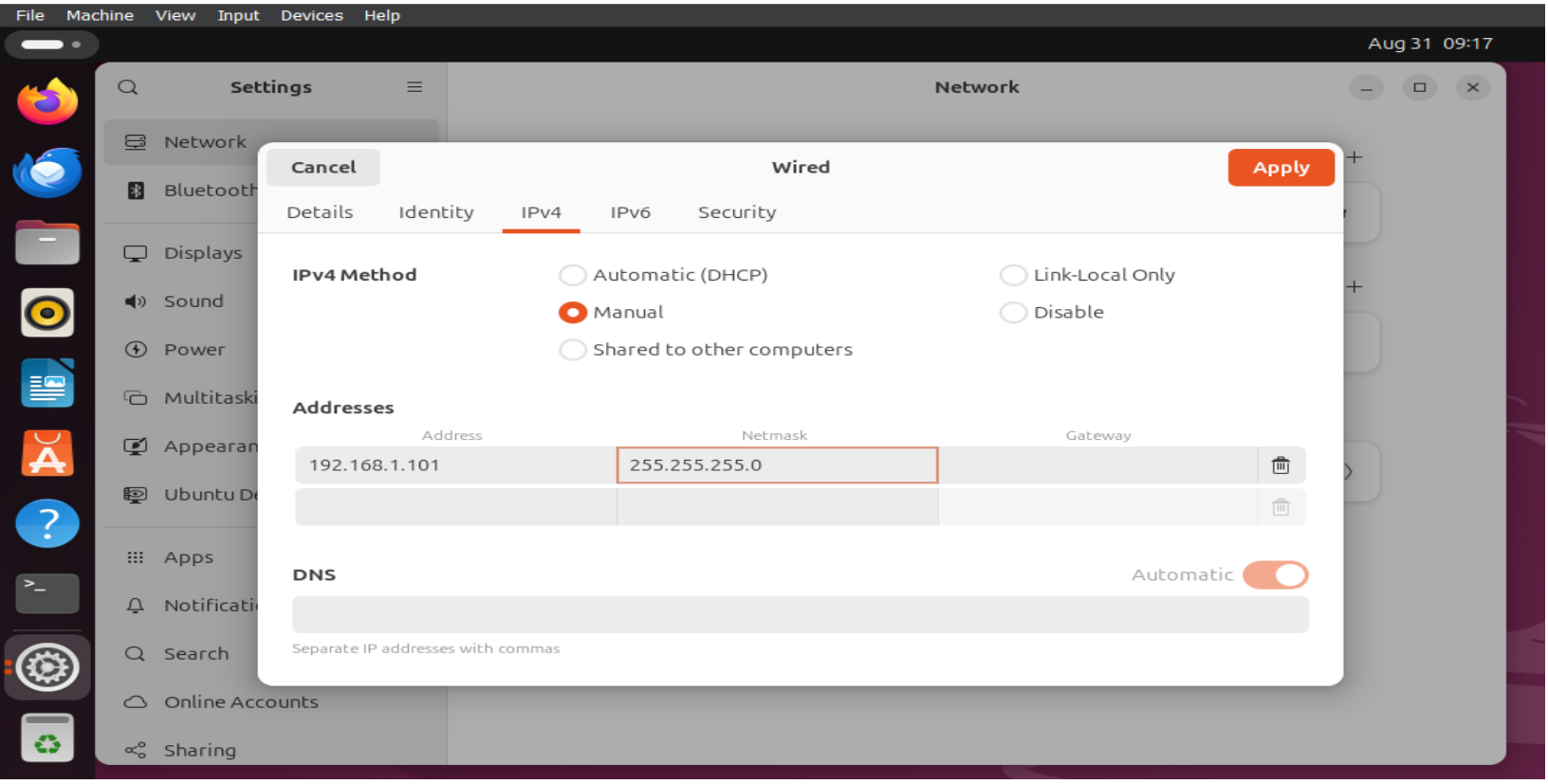




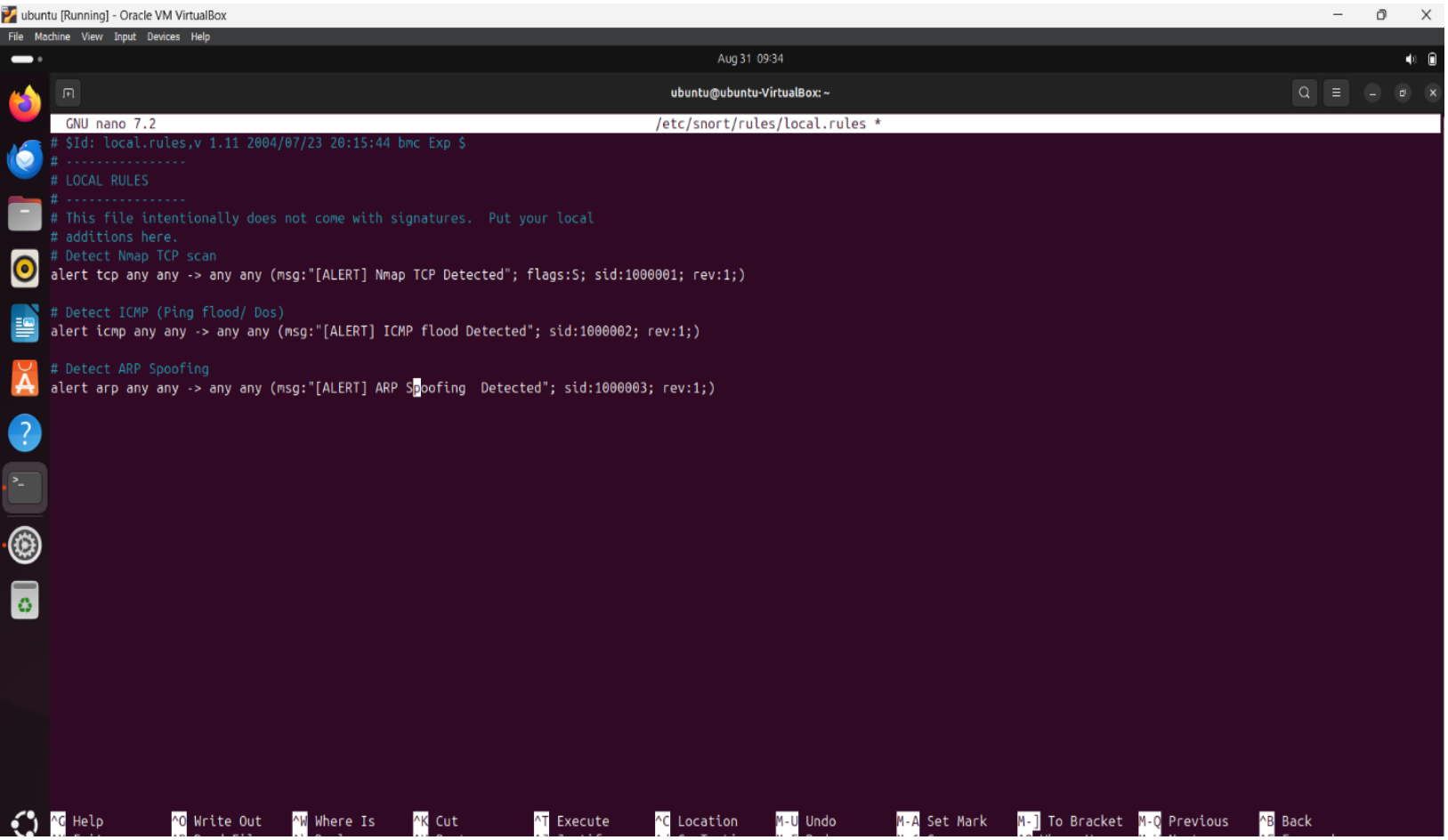
I changed kali (attack vm) network adapter to host-only adapter and also ubuntu network adapter, so that both vm's can communicate



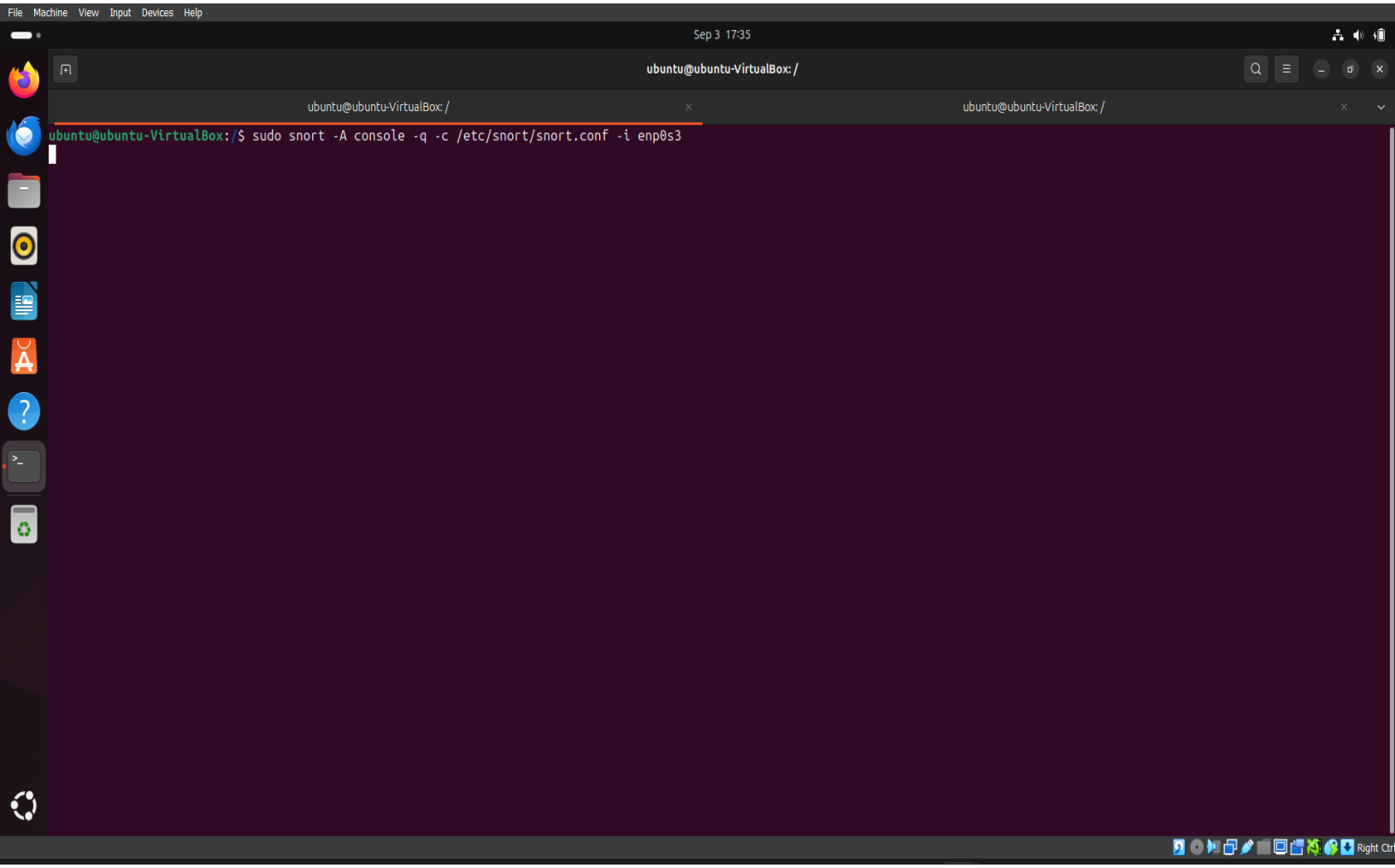
Then I configured kali network by assigning an ip address 0f 192.168.1.100 and a netmask of 255.255.255.0



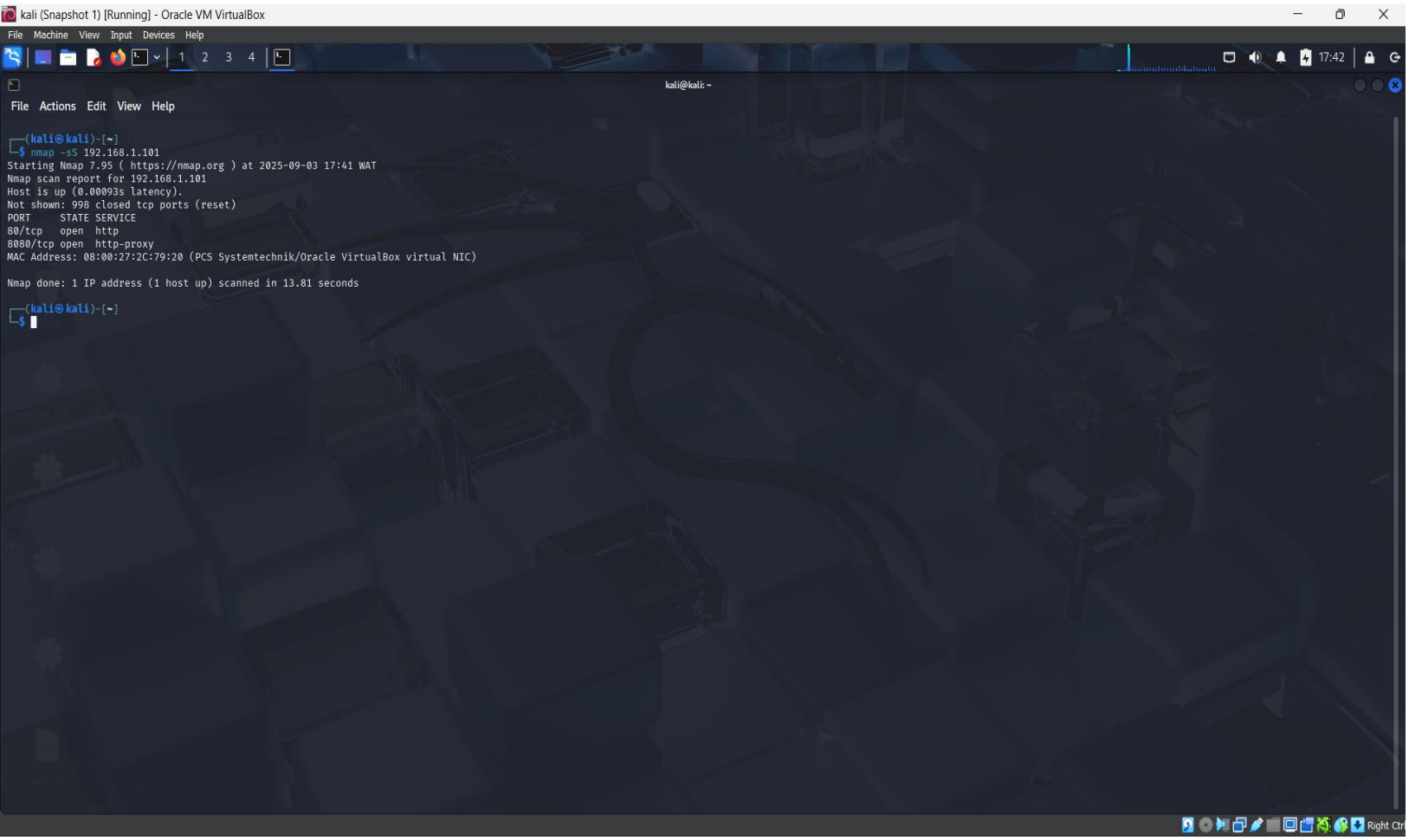
and I configured ubuntu network also by assigning an ip address 0f 192.168.1.101 and a netmask of 255.255.255.0



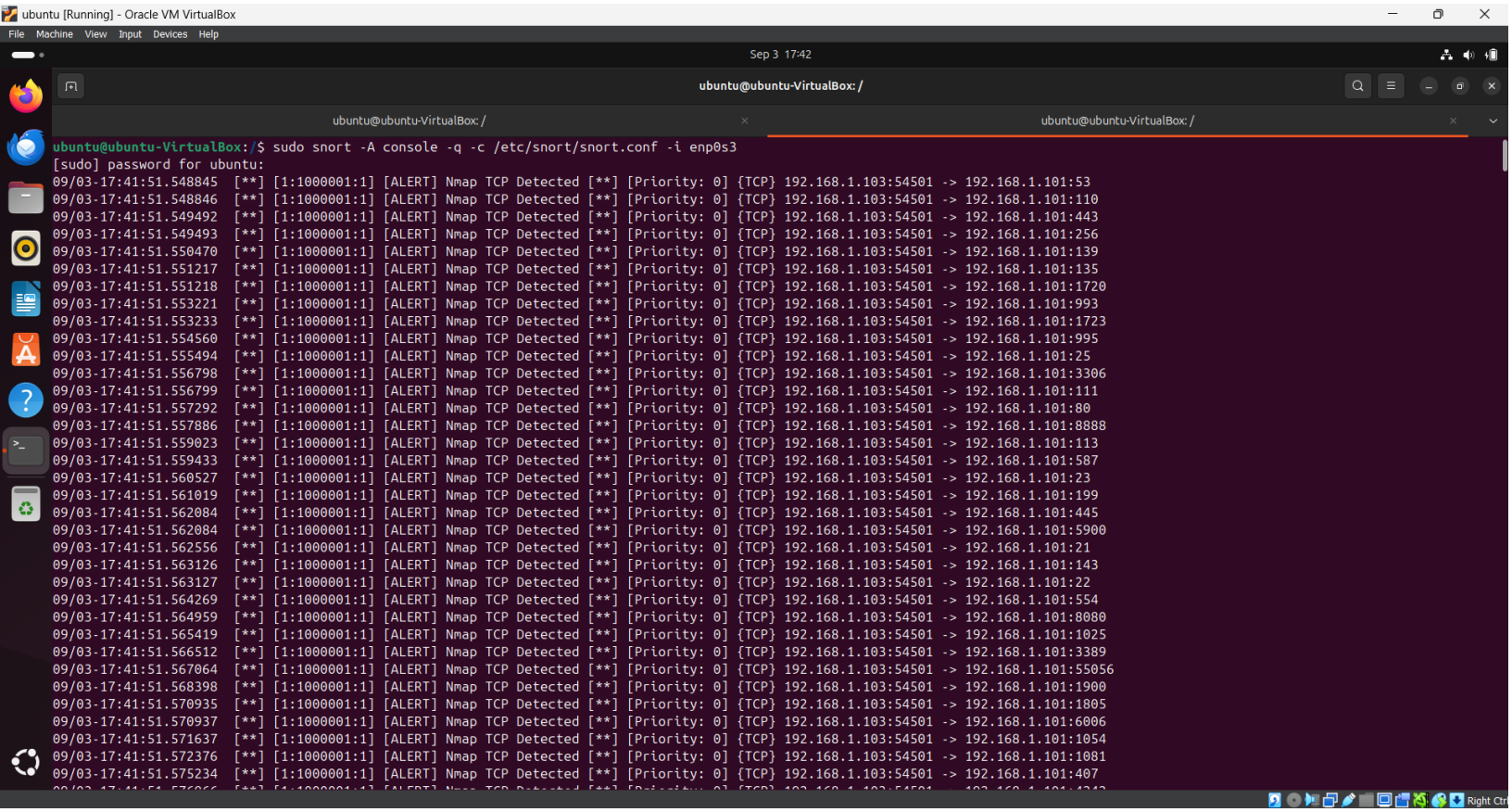
I set the rule on snort to detect tcp scan, icmp flood and arp spoofing



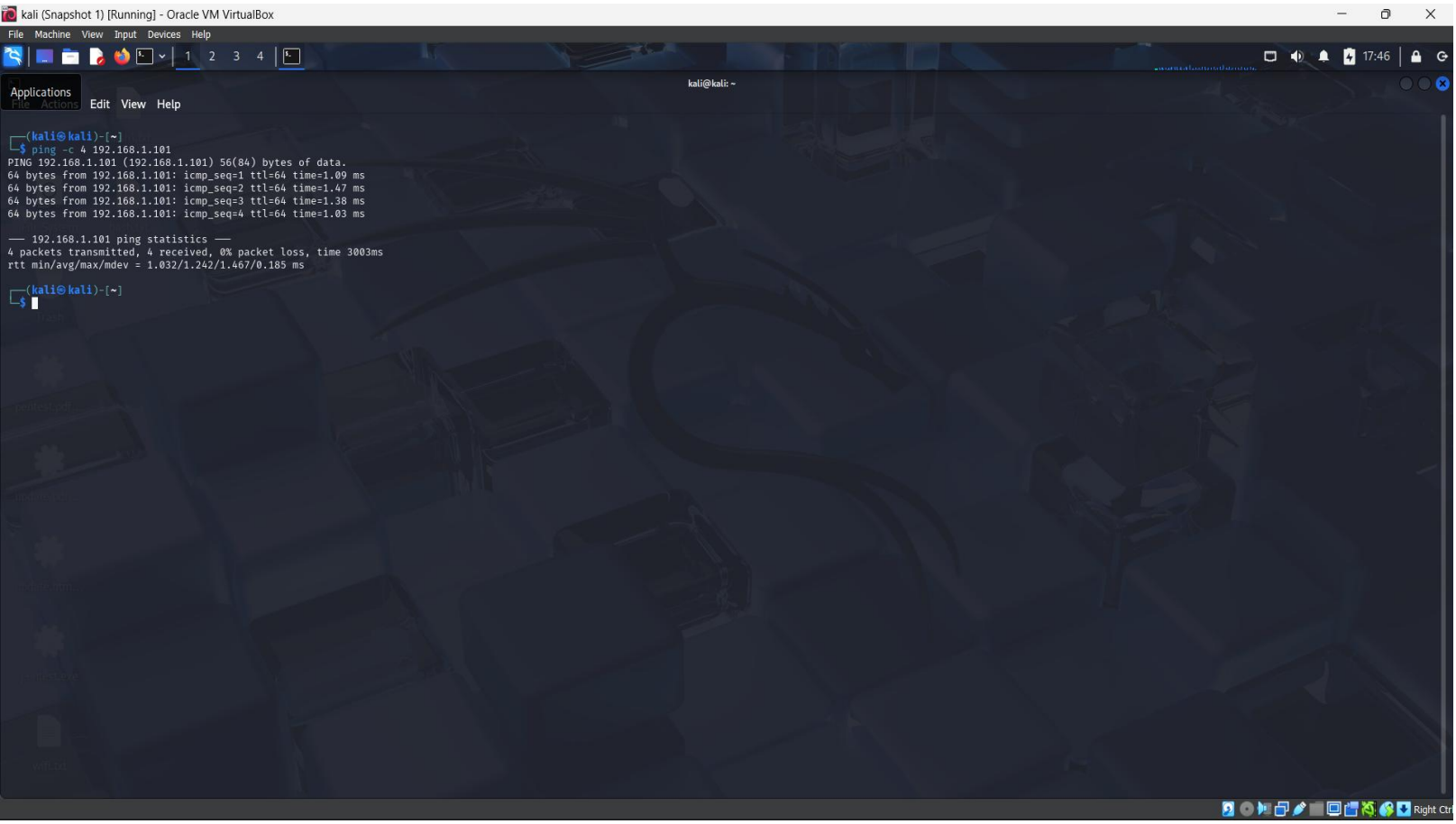
then I started snort using the following command: sudo snort -A console -q -c /etc/snort/snort.conf -i eth1



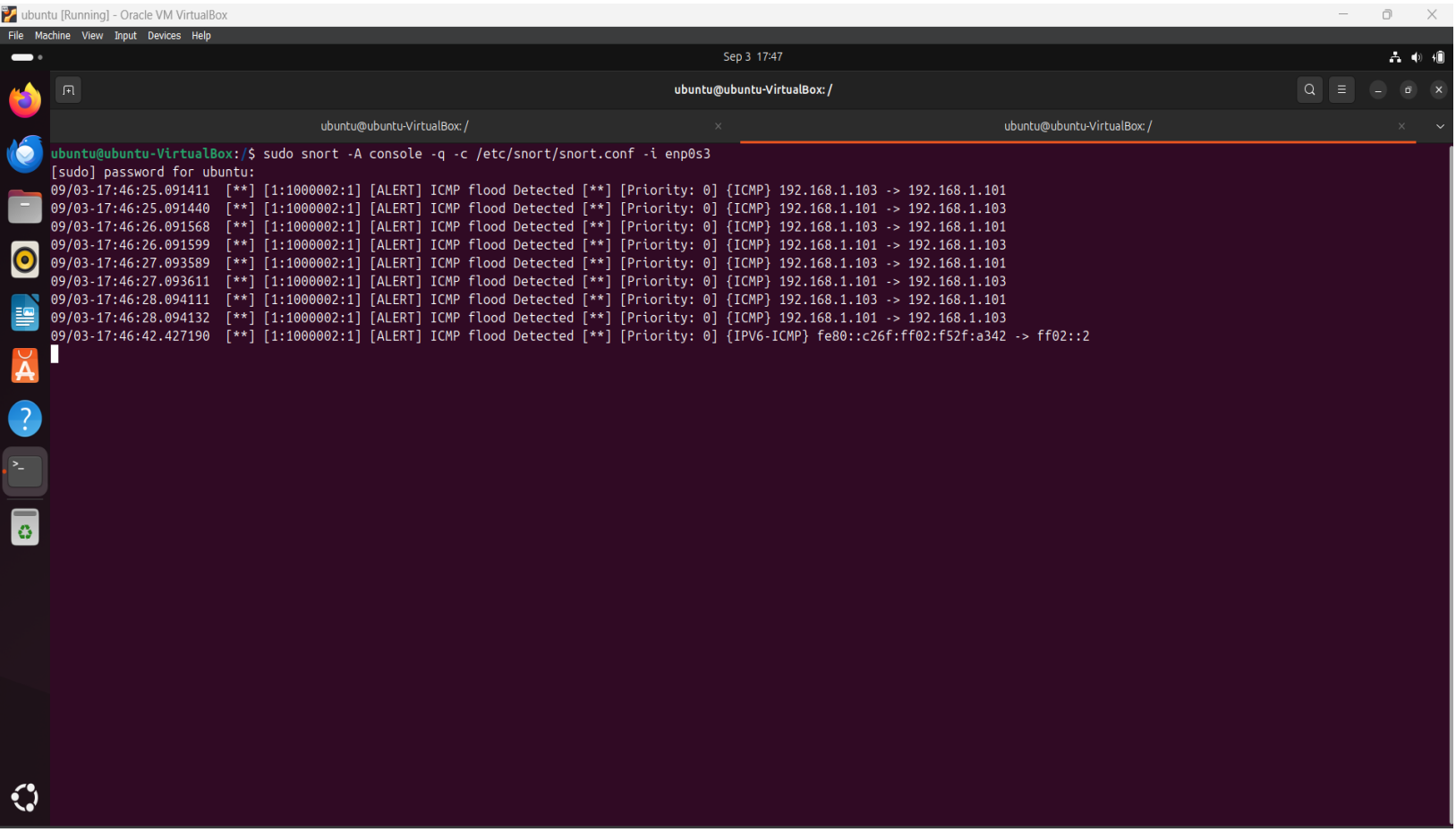
I attempted scanning the target ip address using: nmap -sS 192.168.1.101



The tcp rule was triggered: alert tcp any any -> any any (msg:"[ALERT] Nmap TCP Scan Detected"; flags:S; sid:1000001; rev:1;)



I pinged the victim ip address: ping -c 4 192.168.1.101



And the icmp rule was triggered: alert icmp any any -> any any (msg:"[ALERT] ICMP Flood Detected"; sid:1000002; rev:1;)

Mitigation Recommendation

- Enable port security & DHCP snooping on switches to reduce spoofing
- Deploy IDS in-line
- Apply firewall rules to mitigate Dos (Denial of service)
- Use VLAN segmentation to limit attack surface